

PRACTICAL NO 3

GITHUB LINK

Aim:

(A)Linux Network Analysis

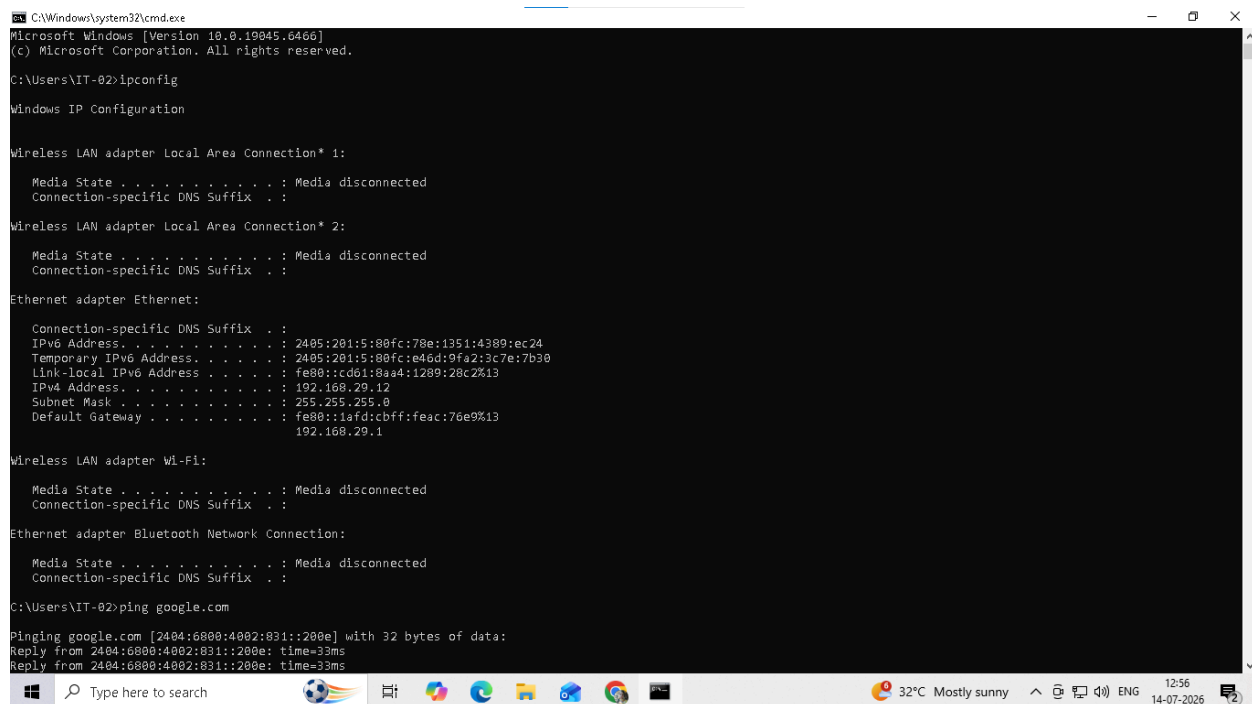
- Linux Network Analysis:

Steps: COMMAND:

1. ipconfig

Description:

Displays the network configuration of the computer, such as the IP address, Subnet Mask, and Default Gateway.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
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C:\Users\IT-02>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2405:201:5:80fc:70e:1351:4300:ec24
    Temporary IPv6 Address. . . . . : 2405:201:5:80fc:e46d:9fa2:3c7e:7b30
    Link-local IPv6 Address . . . . . : fe80::cd61:8aa4:1209:20c2%13
    IPv4 Address. . . . . : 192.168.29.12
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1afd:cbff:feac:70e9%13
                                192.168.29.1

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\IT-02>ping google.com

Pinging google.com [2404:6800:4002:831::200e] with 32 bytes of data:
Reply from 2404:6800:4002:831::200e: time=33ms
Reply from 2404:6800:4002:831::200e: time=33ms
```

2. ping google.com

Description:

Tests the network connectivity between the computer and a remote host by sending ICMP echo requests and measuring the response time.

```
C:\Users\IT-02>ping google.com

Pinging google.com [2404:6800:4002:831::200e] with 32 bytes of data:
Reply from 2404:6800:4002:831::200e: time=33ms
Reply from 2404:6800:4002:831::200e: time=33ms
Reply from 2404:6800:4002:831::200e: time=33ms
Reply from 2404:6800:4002:831::200e: time=31ms

Ping statistics for 2404:6800:4002:831::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 31ms, Maximum = 33ms, Average = 32ms
```

3. netstat

Description:

Displays active network connections, listening ports, and network statistics of the system.

```
C:\Users\IT-02>netstat

Active Connections

 Proto Local Address           Foreign Address         State
TCP    192.168.29.12:7680      192.168.29.116:50002    TIME_WAIT
TCP    192.168.29.12:50041     ymq01r9b:https         CLOSE_WAIT
TCP    192.168.29.12:50163     lb-140-82-113-26-iad:https ESTABLISHED
TCP    192.168.29.12:50289     4.198.108.40:https      CLOSE_WAIT
TCP    192.168.29.12:50301     52.140.232.67:https     ESTABLISHED
TCP    192.168.29.12:50302     192.168.29.245:ms-do    SYN_SENT
TCP    [2405:201:5:80fc:e46d:9fa2:3c7e:7b30]:49687 [2603:1040:a06:6::]:https ESTABLISHED
TCP    [2405:201:5:80fc:e46d:9fa2:3c7e:7b30]:50044 lcbomp-in-f188:5228     ESTABLISHED
TCP    [2405:201:5:80fc:e46d:9fa2:3c7e:7b30]:50238 [2606:4700:8d72:7cbc:c2d7:dff:5ff2:75c4]:https ESTABLISHED
TCP    [2405:201:5:80fc:e46d:9fa2:3c7e:7b30]:50254 [2606:4700:8d71:d8b7:9a16:6e:ccf5:ac50]:https ESTABLISHED
TCP    [2405:201:5:80fc:e46d:9fa2:3c7e:7b30]:50258 [2606:4700:8d73:c0c9:e8d7:dfe:1768:85e2]:https TIME_WAIT
TCP    [2405:201:5:80fc:e46d:9fa2:3c7e:7b30]:50282 [2405:200:1602::312c:823e]:https CLOSE_WAIT
```

4. tracert google.com (Windows) / traceroute google.com (Linux)

Description:

Shows the route (hops) that packets take from the computer to the destination host.

```
C:\Users\IT-02>tracert google.com
```

```
Tracing route to google.com [2404:6800:4002:831::200e]  
over a maximum of 30 hops:
```

1	<1 ms	<1 ms	<1 ms	2405:201:5:80fc:1afd:cbff:feac:76e9
2	*	*	*	Request timed out.
3	7 ms	27 ms	3 ms	2405:203:400:100:172:31:2:22
4	4 ms	2 ms	3 ms	2405:200:801:c00::1240
5	3 ms	4 ms	4 ms	2405:200:803:3168:61::6
6	*	*	*	Request timed out.
7	4 ms	5 ms	3 ms	2405:200:801:c00::1234
8	*	*	*	Request timed out.
9	4 ms	5 ms	4 ms	2001:4860:1:1::39ba
10	2 ms	2 ms	4 ms	2001:4860:1:1::39ba
11	4 ms	5 ms	7 ms	2001:4860:0:1::87f1
12	17 ms	9 ms	10 ms	2001:4860:0:1::9de8
13	*	*	*	Request timed out.
14	38 ms	32 ms	34 ms	2001:4860::9:4001:d9e7
15	35 ms	34 ms	34 ms	2001:4860:0:1::539b
16	33 ms	34 ms	33 ms	tzdelb-bj-in-x0e.1e100.net [2404:6800:4002:831::200e]

```
Trace complete.
```